

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres

One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Hints. Read them progressively and stop as soon as one is enough to restart the problem.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that $I = (x) \cap (y, z)$.
- (ii) Determine the irreducible components and their generic points.
- (iii) Is X reduced, connected, irreducible, or integral?

Hint 1 (conceptual). For an affine scheme, generic points of irreducible components correspond to minimal prime ideals.

Hint 2 (technical). To prove $(x) \cap (y, z) = (xy, xz)$, start with $f = xg$ in the intersection and reduce the equality modulo (y, z) .

Hint 3 (conceptual). An intersection of prime ideals is radical. For connectedness, decide whether the two components meet; for irreducibility, return to the definition.

Exercise 2. The function field of the cusp

★★ CORE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral, compute its function field, and deduce its dimension.

Suggested strategy. Embed the coordinate ring in $\mathbb{k}[t]$, prove that the embedding is injective, and then compare fraction fields.

Hint 1 (technical). Reduce every class modulo $y^2 - x^3$ to $A(x) + yB(x)$. After substituting $x = t^2$, $y = t^3$, separate even and odd powers of t .

Hint 2 (technical). The fraction field contains y/x . What does this element become in $\mathbb{k}(t)$?

Exercise 3. Chains and dimensions

★ CORE

- (i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.
- (ii) Give a chain of length n in $\mathbb{A}_{\mathbb{k}}^n$.
- (iii) Determine the dimension of

$$\operatorname{Spec}(\mathbb{k}[x, y, z]/(xy, xz))$$

from its components.

Hint 1 (conceptual). Chains of irreducible closed subsets correspond, with inclusions reversed, to chains of prime ideals.

Hint 2 (technical). In affine space, successively impose $x_1 = 0$, then $x_2 = 0$, and so on. For the last scheme, take the maximum of the dimensions of its irreducible components.

Exercise 4. Noether normalisation and dimension

★★ CORE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x^3).$$

- (i) Show that B is finite over $\mathbb{k}[x]$.
- (ii) Use this to recover $\dim(B) = 1$.

(iii) Compare with the computation from the function field.

Hint 1 (technical). The element y satisfies a monic polynomial over $\mathbb{k}[x]$. Use division by that monic polynomial to generate B as a $\mathbb{k}[x]$ -module.

Hint 2 (conceptual). Recall that a finite integral extension preserves Krull dimension. Compare the result with $\text{trdeg}_{\mathbb{k}} K(C)$.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line

★ **CORE**

Let

$$\pi : \mathbb{A}_{\mathbb{C}}^2 = \text{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}_{\mathbb{C}}^1 = \text{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t - a)$ and its fibre over the generic point (0) . Explain why the latter is $\text{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Hint 1 (conceptual). The fibre over a point $s \in \text{Spec}(A)$ is obtained by tensoring with its residue field $\kappa(s)$.

Hint 2 (technical). For the closed point $(t - a)$, tensor with $\mathbb{C}[t]/(t - a)$. For the generic point (0) , tensor with $\kappa((0)) = \mathbb{C}(t)$.

Exercise 6. A fibre product larger than the set-theoretic one

★★ **CHOICE**

Let $X = \text{Spec}(\mathbb{C})$ and $S = \text{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Hint 1 (technical). For affine schemes, compute the tensor product $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$. Present the first copy as $\mathbb{R}[T]/(T^2 + 1)$ and then extend scalars to $\mathbb{C}$.

Hint 2 (technical). Factor $T^2 + 1$ over $\mathbb{C}$ and apply the Chinese remainder theorem.

Exercise 7. The power map and its fibres

★★ **CORE**

Assume that $\mathbb{k}$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^1, \quad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Suggested strategy. Write the coordinate-ring map using a target coordinate s . Compute closed fibres by imposing $s = a$, then treat the generic point by replacing $\mathbb{k}[s]$ with $\mathbb{k}(s)$.

Hint 1 (technical). The fibre over a has coordinate ring $\mathbb{k}[t]/(t^n - a)$. Separate $a = 0$ from $a \neq 0$ and use the derivative of $t^n - a$ to test repeated roots.

Hint 2 (technical). For the generic fibre, study $t^n - s$ over $\mathbb{k}(s)$. Eisenstein at the prime s in $\mathbb{k}[s]$ proves irreducibility.

Hint 3 (technical). To prove finiteness, divide exponents modulo n : express each polynomial uniquely as

$$\sum_{r=0}^{n-1} t^r a_r(t^n).$$

Exercise 8. A family whose special fibre becomes reducible

★★ **CORE**

Let

$$X = \text{Spec}(\mathbb{k}[x, y, t]/(xy - t)) \longrightarrow \mathbb{A}_{\mathbb{k}}^1.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Hint 1 (technical). The fibre over a is obtained by setting $t = a$, hence has equation $xy = a$.

Hint 2 (technical). If $a \neq 0$, one coordinate is invertible and the ring is a Laurent polynomial ring. If $a = 0$, factor the equation and identify the two components.

Hint 3 (conceptual). Irreducibility can change in a family even when the fibre dimension remains constant.

Exercise 9. A jumping fibre★★ **CORE**

Consider

$$f : \mathbb{A}_{\mathbb{k}}^2 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad (x, y) \longmapsto (x, xy).$$

Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Suggested strategy. Write the fibre ring over (a, b) , eliminate x using $x - a$, and then distinguish the cases $a \neq 0$, $a = 0, b \neq 0$, and $(a, b) = (0, 0)$.

Hint 1 (technical). The fibre ring is

$$\mathbb{k}[x, y]/(x - a, xy - b).$$

After substituting $x = a$, the remaining equation is $ay = b$.

Hint 2 (technical). For the generic fibre, work over $\mathbb{k}(u, v)$: the relation $x = u$ makes x invertible and determines y .

Exercise 10. The generic fibre formula in examples★★ **CHOICE**

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}_{\mathbb{k}}^{m+n} \rightarrow \mathbb{A}_{\mathbb{k}}^m$.

Hint 1 (conceptual). Compute the generic fibre using function fields. Its dimension is the transcendence degree of $K(X)$ over $K(Y)$.

Hint 2 (technical). For $(x, y) \mapsto (x, xy)$, recover x and y rationally from the target coordinates $u = x$ and $v = xy$.

Optional outreach exercises: Hilbert schemes of points**Exercise 11. $\text{Hilb}^n(\mathbb{A}^1)$** ★★ **CHOICE**

Assume that $\mathbb{k}$ is algebraically closed. Show that every ideal $I \subset \mathbb{k}[x]$ with $\dim_{\mathbb{k}} \mathbb{k}[x]/I = n$ is generated by a unique monic polynomial of degree n . Deduce

$$\text{Hilb}^n(\mathbb{A}_{\mathbb{k}}^1) \simeq \mathbb{A}_{\mathbb{k}}^n$$

and explain how factorisation records support and multiplicities.

Suggested strategy. First classify finite-colength ideals of the PID $\mathbb{k}[x]$. Then identify a monic degree- n polynomial with its n coefficients. Keep the functorial argument separate from the classification of $\mathbb{k}$ -points.

Hint 1 (technical). If $I = (f)$ with f monic of degree d , Euclidean division gives the basis $1, x, \dots, x^{d-1}$ of $\mathbb{k}[x]/(f)$.

Hint 2 (conceptual). For the scheme-level statement over a ring R , multiplication by x on the rank- n quotient satisfies its characteristic polynomial by Cayley–Hamilton.

Hint 3 (technical). Over an algebraically closed field, factor the monic polynomial. Distinct roots give the support and their exponents give the local lengths.

Exercise 12. Commuting matrices and a cyclic vector★★★ **OPTIONAL**

Let V be n -dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\mathbb{k}[X, Y]v = V$. Show that

$$I = \{f \in \mathbb{k}[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and $\mathbb{k}[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.

Suggested strategy. Use the commuting pair to make V into a $\mathbb{k}[x, y]$ -module, with x acting by X and y by Y . The cyclic vector then defines a surjective module map from $\mathbb{k}[x, y]$ to V .

Hint 1 (technical). Commutativity is exactly what makes $f \mapsto f(X, Y)$ an algebra homomorphism. Use it to show that the annihilator of v is an ideal.

Hint 2 (technical). For $I = (y, x^2)$, use the basis $(1, x)$ of the quotient and compute multiplication by x and by y in this basis.